

Buoyancy fluxes of sea-ice leads in the Arctic

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Key Points:

- ideas

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Abstract

Plain Language Summary

1 Introduction

- Discuss Sea ice and insulator properties.
- Describe leads, their scales, their formation mechanisms + climatology.
- Describe believed importance in the fluxes.
- Describe Effect of leads in the surrounding ice pack.

2 Methods

Model description

LKF identification

Refer to Hutter, N., et. al. (2019)

- Describe MLD definition
- Describe buoyancy flux and decomposition between thermal and haline components.

3 Results

Figures:

1. Satellite and model image of LKF
2. Buoyancy flux on 2 different snapshots and LKF mask (Total deformation panel?).
3. Time-series of area cover + buoyancy fluxes. (Merge the 0-80
4. Water mass transformation of the Pack Ice and the LKF.
5. Density composite for summer and winter? (Potential to affect interior)

3.1 Observation vs Models

Figure 1

- Describe spatial pattern of features, for example in satellite we observe many floes of different scales that the model can not reproduce
- Describe that the model is a mean state over a day, thus it is likely smoother.
- Highlight that model still reproduces features similar to those observed in the model.
- The length-scales also appear to match (Should we do something to quantify this?)

3.2 Impact of LKF in the ocean buoyancy

Figure 2

- Describe the sign of the buoyancy flux (Negave = denser water).

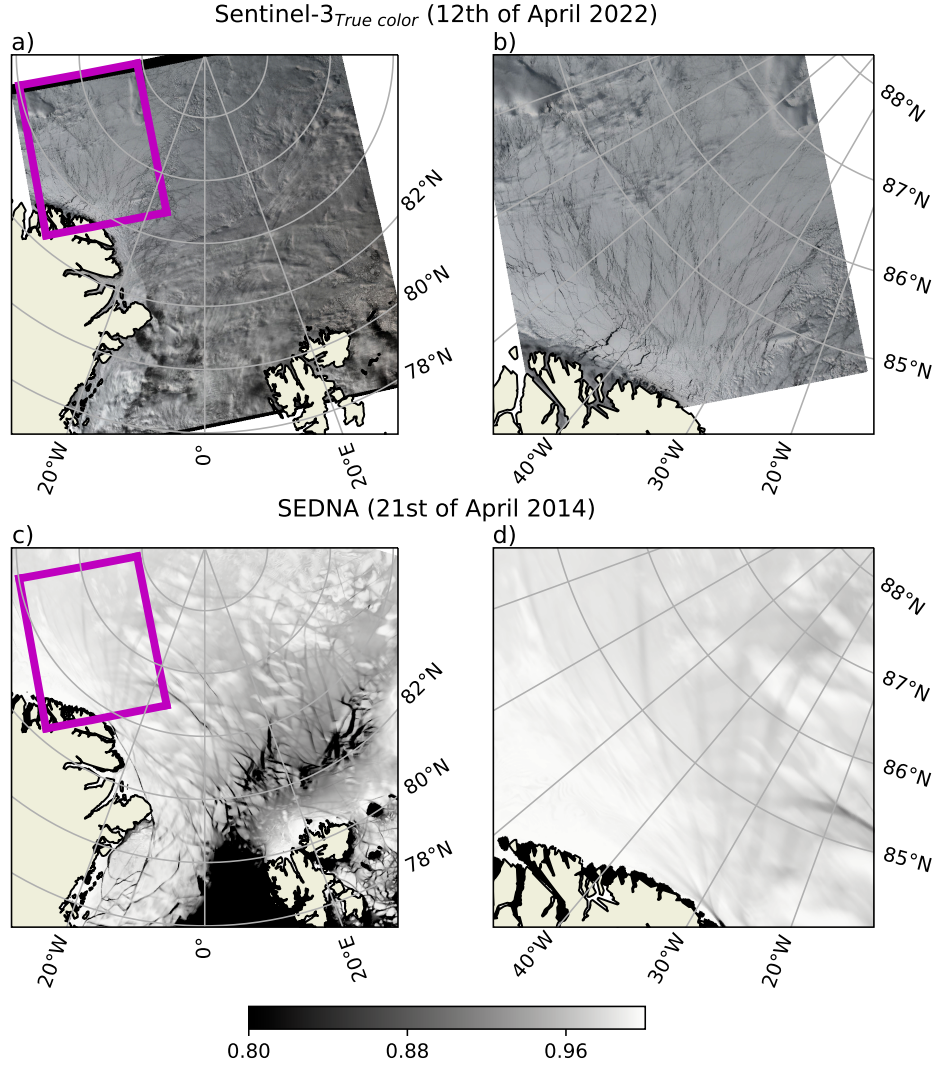


Figure 1. Linear kinematic features from observations and high resolution model. a) True color from Sentinel 3 on the 12th of April 2022. b) Zoom of magenta area in panel a) showing the linear kinematic features on the 12th of April 2022. c) Mean ice concentration from SEDNA (800m resolution) on the 21st of April 2014. d) Zoom of magenta area in panel d) showing the linear kinematic features on the 21st of April 2014.

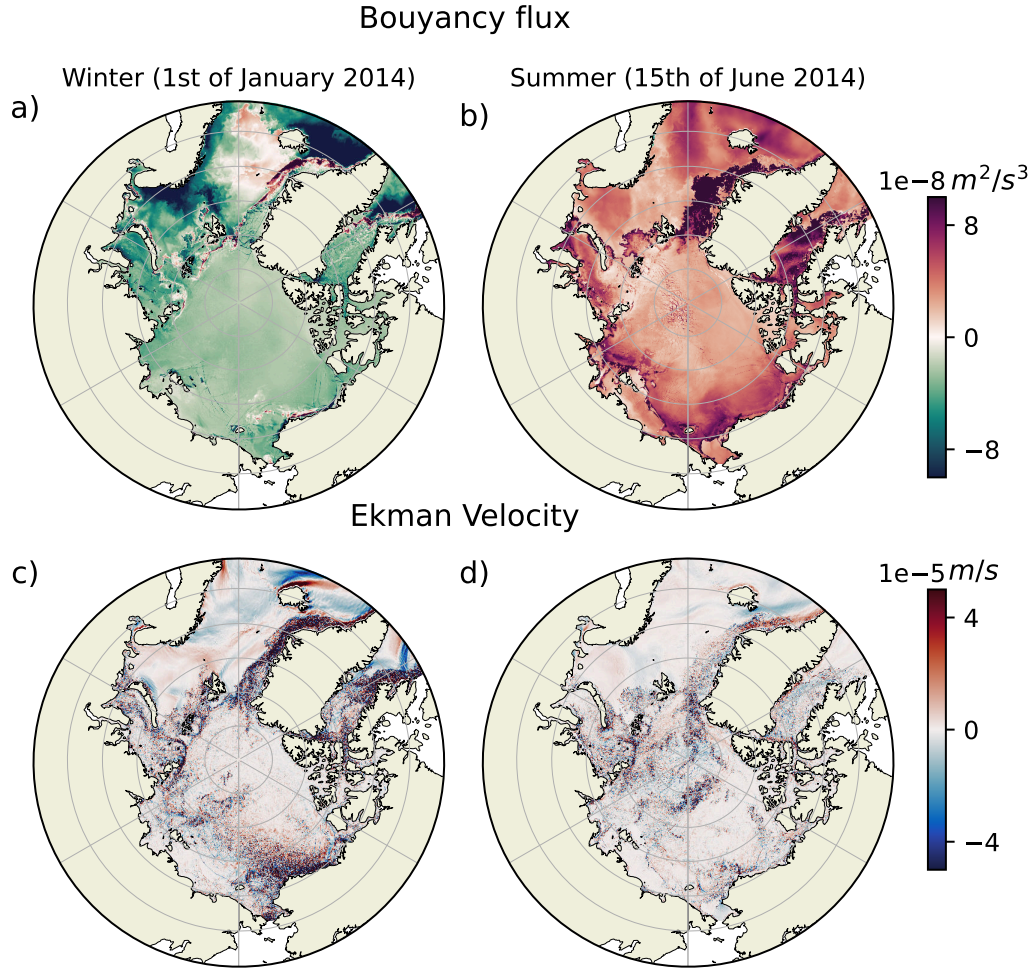


Figure 2. Bouyancy flux and Ekman Velocity for two snapshots during winter and summer. a) Bouyancy flux in Winter the 1st of January 2014. b) Bouyancy flux in Summer the 15th of June 2014. c) Ekman velocity in winter the 1st of January 2014. d) Bouyancy flux in summer the 15th of June 2014

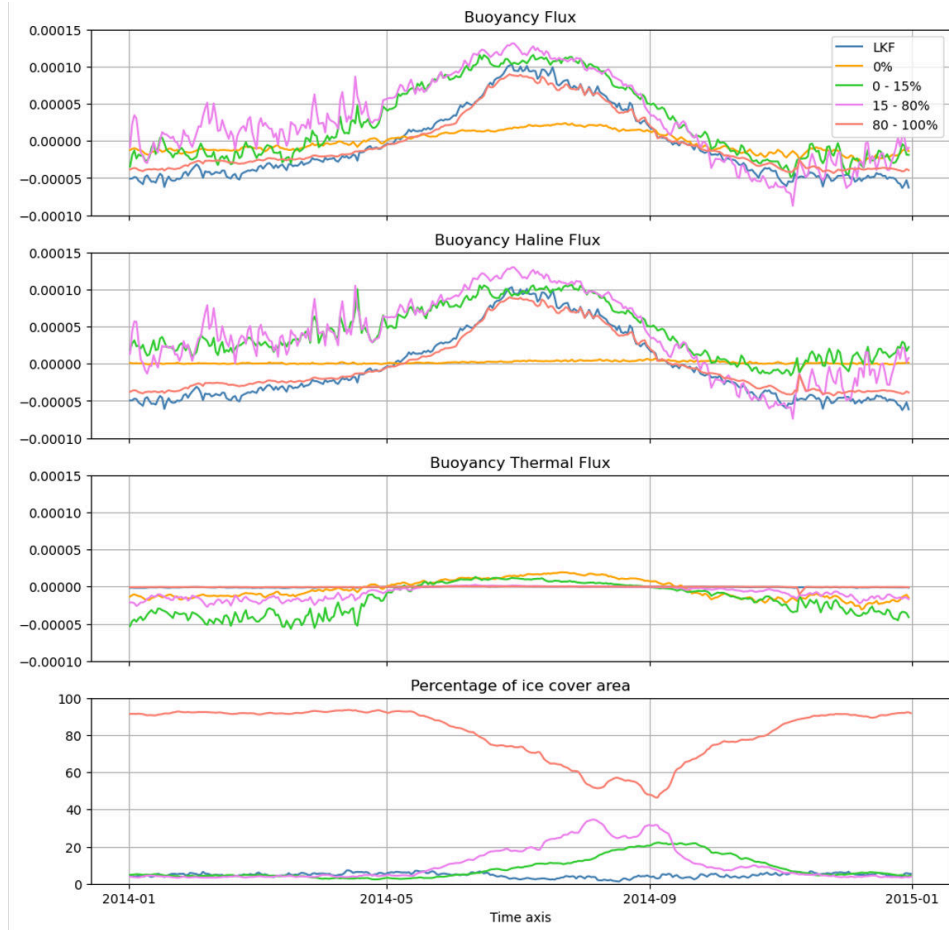


Figure 3. Buoyancy flux and Ekman velocities normalized by the ice cover area of linear kinematic features, transitional ice zone (ice concentration 0-80%), and pack ice (ice concentration >80%).

- Describe that we can see important differences in the BF between the open ocean and the packed ice.
- Note some of the LKF features observed in the buoyancy flux, for example large negative flux in the East Siberian Sea or Beaufort sea and the positive flux in the center of the Arctic.
- Describe analogously the Ekman velocities, negative = Ekman Pumping = downwelling.
- In the regions where we see larger buoyancy flux, we also observe large Ekman velocities.
- Talk about how the Ekman velocity depends on the atmospheric forcing.

Figure 3

- This figure will be updated
- TODO: Remove 0% concentration, merge 0-80% concentration.
- TODO: Add time series of Ekman Velocity. Remove the Thermal component of BF.

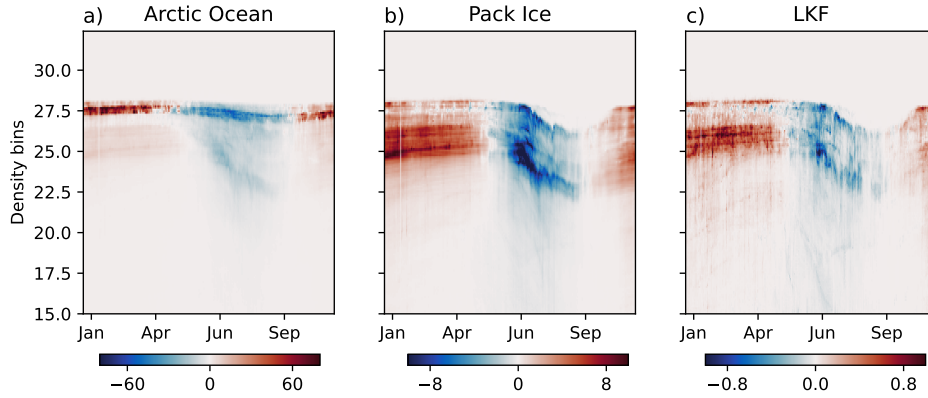


Figure 4. Hovmoller diagram of the water mass transformation (sv). Water mass transformation of for a) the Arctic ocean, b) pack ice, c) and linear kinematic features. Note the changes in the magnitude of the colorbar of each panel.

- Describe that the Buoyancy flux is dominated by a seasonal cycle with small high frequency variability.
- Important seasonal cycle, buoyant water in Summer, denser in Winter.
- Describe that area in LKF is really small $< 5km$.
- Fluxes are mostly dominated by haline component
- Fluxes in LKF are comparable to those in the Pack ice.
- Explain that despite the area being just a fraction, generally less than 10% of the ice-covered Arctic, the fluxes in LKF explain the fluxes of the packed ice and their magnitude is comparable to the marginal ice zone.

Figure 4

- TODO: Add units below colorbar.
- Describe seasonal cycle of water mass transformation
- Highlight the change in the magnitude
- The LKF explain around 10% of the Pack Ice WMT.
- During summer the transformation is towards buoyant water and in winter the transformation is towards denser water.
- Describe shift of water mass transformation during summer.
- Is it possible to estimate the effect of water masses transformation based on the outcrop of density contours in the Arctic?

3.3 Impact of LKF in the ocean interior

Figure 5

- The composites shown important signature of buoyant and fresher water near adjacent to the LKF.
- The lense of buoyant water appears to vary with the season, but more info is needed to conclude.
- The average response near the LKF appears to be the same independent to the season but with an intensification in summer (melting in general in LKFs?), more needs to be done to conclude.

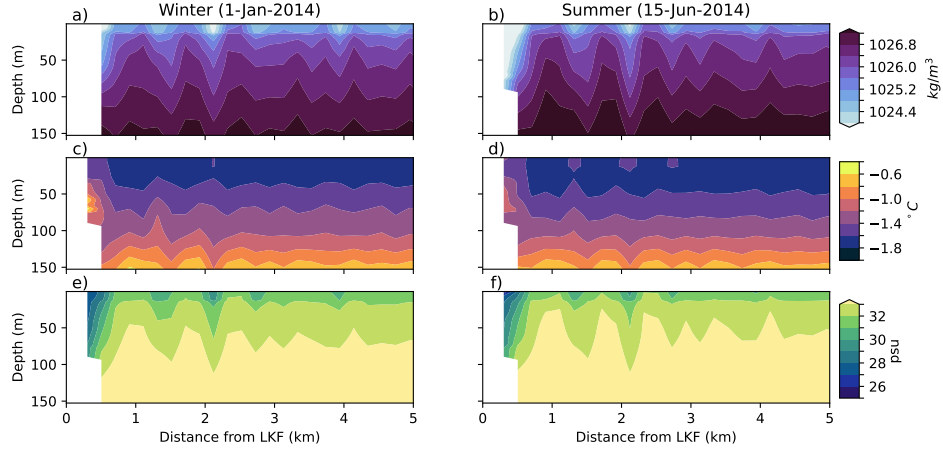


Figure 5. Composite of density, temperature, and salinity normal to the identified linear kinematic features on the 1st of January and 15th of June 2014. a) Density in Winter the 1st of January 2014. b) Density in Summer the 15th of June 2014. c) Temperature in winter the 1st of January 2014. d) Temperature in summer the 15th of June 2014. e) Salinity in winter the 1st of January 2014. f) Salinity in summer the 15th of June 2014

- We observe other lenses of buoyant and fresher water, likely linked to past LKF events.
- Tilting of isopycnals near the LKF?
- Warmer water near the LKF? connection between the intermediate water and the surface through LKF?
- What is the role of mixing here? since the water near the LKF is really buoyant?

4 Conclusions

5 Open Research

Acknowledgments

References